Exact Context-Edit Stability for Plurality-Based Functional-Dependency Checks

Trần Vũ Bảo Minh

Independent Researcher

Revised working preprint – November 5, 2026

Abstract

A plurality-based functional-dependency check judges a target value using rows that match its determinant values. When determinant cells can change, group membership changes as well. We study the minimum number of context-cell edits that reverses a fixed target's decision, protecting its entire row. For checks with a common dependent attribute and independent unit-cost categorical edits, we prove a row-action dominance reduction for making a rejected target accept. For the opposite direction, we derive an exact formula: nonsingleton groups retain their ordinary vote-margin cost, and only singleton groups can make editable keys strictly reduce the global radius. Exhaustive finite checks and a bounded optimization comparison support the implementations. A reused-source study covers 18,734 target cells from Beers, Flights and Rayyan. Among 13,762 accepted targets, 1,133 fixed-key infinite radii become finite and one finite radius decreases. No strict reduction appears in 30 exact rejected-target pairs; two full models time out. Stability does not establish correctness: 1,493 accepted targets disagree with references despite radius at least two.

1 Introduction

Functional dependencies (FDs) describe agreement constraints between attributes. In an imperfect table, a simple detector can compare an observed dependent value with the unique plurality among rows sharing its determinant values. Such a decision depends on context: changing another row's key can move that row into or out of the target's group. A certificate that protects all keys therefore answers a different question from one allowing key errors.

We study this distinction as an exact decision-sensitivity problem. The target value is not corrected, the target row is immutable, and the edited table need not satisfy all FDs. We count changed cells in other rows, rather than insertions, deletions, changed records, or changes to an output label. A radius certifies stability under its stated edit model; it does not certify factual validity.

Our technical results concern several checks sharing one dependent attribute. First, a componentwise dominance lemma reduces accepting counterfactuals to at most $2^q + 1$ actions per context row, where q is the number of key columns. Second, an exact rejection formula identifies a singleton bottleneck: keys strictly lower an accepted target's radius exactly when a singleton group's cheapest rival import costs less than every nonsingleton group's margin cost. This gives different algorithms for the two decision directions.

Minimum interventions, predicate-equivalent domains, voting margins and integer optimization are established ingredients; we do not claim them as inventions. The contribution presented here is

the specialized dominance and singleton characterization under explicit assumptions, with bounded computational and source-table evidence. In particular, our source results do not show a general accepting-direction advantage from mutable keys.

The intended use is an edit-budget sensitivity audit, not a repair prescription. For a declared context-edit budget B, a radius greater than B certifies that the present decision survives every permitted edit within that budget; a radius at most B supplies an explicit counterexample to that certificate. This lets an auditor distinguish trusting determinant cells from allowing them to be wrong. It does not establish which cells are actually wrong, or validate the choice of budget and unit costs. We position the result as a restricted theory/algorithm note, rather than an accuracy-improving cleaning system.

2 Model and edit semantics

Let $T = (t, r_1, \ldots, r_{n-1})$ be identified rows with $q \geq 1$ categorical key attributes and one dependent attribute y. The finite RHS domain has $d \geq 2$ values. Each key domain contains the target value and an alternative. By independently relabeling column values, assume $t = (0, \ldots, 0)$. Rules have nonempty determinants $X_g \subseteq \{1, \ldots, q\}$, $g = 1, \ldots, m$, and common RHS $y \notin X_g$. Define

$$G_g(T) = \{r \in T : r[X_g] = t[X_g]\}, \qquad c_{gv}(T) = |\{r \in G_g(T) : r_y = v\}|.$$

The detector accepts the target iff

$$D(T, t) = 1 \quad \iff \quad c_{g0}(T) > c_{gv}(T) \quad \text{for all } g,\ v \neq 0. \tag{1}$$

Ties reject. Unique plurality does not require an absolute majority when $d > 2$.

An edited table T' has the same identified rows, and t is unchanged in every column. Context cells may independently take any value of their column's domain. The cost is cell Hamming distance

$$H(T, T') = \sum_{i=1}^{n-1} \sum_{j=1}^{q+1} \mathbf{1}[r_{ij} \neq r'_{ij}].$$

Let ρ_{full} minimize this cost subject to $D(T', t) \neq D(T, t)$. Let ρ_{fixed} impose the additional restriction that all key cells remain unchanged. An impossible flip has radius ∞. Every integer budget $k < \rho$ preserves the initial decision. At cost $\rho < \infty$, a witness flips it.

There are no additional validity constraints, NULL semantics, insertion/deletion, unequal edit costs, or requirements to preserve other targets' decisions. These restrictions are essential to the results below.

A coupled example. Consider the binary table below, with rules $(a, b) \to y$ and $(a, c) \to y$. The first row is protected. Each target group initially has one zero and one one, so both checks reject.

Row	a	b	c	y
Target	0	0	0	0
r_1	0	0	1	1
r_2	0	1	0	1
r_3	1	0	0	0

Table 1: Constructed illustration of a shared key edit, not a source-data result.

With fixed keys, both $r_1.y$ and $r_2.y$ must change to zero: editing either alone leaves the other check tied. Thus $\rho_{\text{fixed}} = 2$. Changing only $r_3.a$ to zero joins both groups simultaneously; each then has two zeros and one one, so $\rho_{\text{full}} = 1$. This example explains why rowwise group membership matters for accepting a rejected target. It does not show that such savings are prevalent in real tables.

3 Making an accepted target reject

Assume $D(T, t) = 1$. Write $a_g = c_{g0}(T)$ and $b_g = \max_{v \neq 0} c_{gv}(T)$, allowing absent values with count zero. For a context row r, let $h_g(r) = |\{j \in X_g : r_j \neq 0\}|$.

Theorem 1 (Exact rejection radius)**.** *Define*

$$C_g = \begin{cases} \lceil (a_g - b_g)/2 \rceil, & |G_g(T)| \geq 2, \\ \min\limits_{r \neq t} \big(h_g(r) + \mathbf{1}[r_y = 0]\big), & G_g(T) = \{t\}, \end{cases}$$

where an empty minimum is ∞. Then $\rho_{\text{full}} = \min_g C_g$.

Proof. For a nonsingleton group, fix a rival v. A key-cell edit changes its target/rival margin by at most one: only one row can enter or leave this group. An RHS-cell edit changes it by at most two. These bounds hold at any intermediate table. A final Hamming edit can be implemented by changing each differing cell once. Rejection therefore costs at least $\lceil (a_g - b_g)/2 \rceil$.

Choose a strongest rival and change that many context RHS-0 cells in the group to it, keeping keys fixed. Each edit reduces the selected margin by two, reaching a tie or loss. There are enough editable zero cells: initial acceptance and a nonsingleton group imply $a_g \geq 2$, and $\lceil (a_g - b_g)/2 \rceil \leq \lceil a_g/2 \rceil \leq a_g - 1$.

For a singleton group, a rejecting destination must contain at least one nonzero context RHS in the target group. That row must change its $h_g(r)$ initially nonmatching keys and also its RHS if originally zero. This lower bound is attained by editing just that row; it then ties the protected target one-to-one. If there is no context row, no flip is possible.

Finally, global rejection is a disjunction of rule rejections. Any flip rejects at least one rule; a minimum-cost witness for any one rule already suffices. Taking the minimum over rules proves the result without an independence assumption about their overlapping groups. □

Corollary 1 (Exact condition for keys to help)**.** *Let A be the minimum nonsingleton margin cost and S the minimum singleton import cost, using ∞ for an absent kind. Then*

$$\rho_{\text{fixed}} = A, \qquad \rho_{\text{full}} = \min(A, S).$$

Thus keys strictly reduce the radius iff $S < A$. A singleton group is necessary, but not sufficient. With no singleton target groups, the radii are equal.

A scan obtains group counts and best singleton imports. After choosing the cheapest rule, it materializes only one witness. For fixed schema and alphabet, this is linear in n; the implementation also scans the RHS alphabet when choosing a rival. The condition is not a claim that singleton values are wrong. For example, with target $(0, 0, 0, 0)$, context $(0, 1, 1, 0)$, and rules $a \to y, b \to y$, a singleton exists but $A = 1 < S = 2$, so both radii are one.

4 Decision distance versus whole-table repair

For comparison within our identified-row model, let $R(T)$ be the minimum context-cell cost making the entire edited table satisfy every FD as an agreement constraint, still protecting t. This comparison protects the same target in both objectives; it is not a redefinition of prior repair semantics.

Proposition 1 (No constant-factor substitution)**.** *Even with one rule $x \to y$ and binary y, the ratios $\rho_{\text{full}}/R(T)$ and $R(T)/\rho_{\text{full}}$ can each be arbitrarily large on instances where both costs are positive and finite.*

Proof. For any $h \geq 1$, use target $(0, 0)$, $2h$ additional $(0, 0)$ rows and one $(0, 1)$ row, with binary domains. Changing the lone one to zero repairs the table; since it initially violates the FD, $R(T) = 1$. Theorem 1 gives $\rho_{\text{full}} = h$ from counts $2h + 1$ and one. Thus the first ratio is h.

For any $k \geq 1$, use target $(0, 0)$ and pairs $(i, 0), (i, 1)$ for $i = 1, \ldots, k$, with key domain $\{0, \ldots, k\}$. Each original conflicting pair requires an edit in one of its own cells; otherwise that pair still violates the FD. The disjoint pairs require at least k edits, and changing each pair's one to zero attains $R(T) = k$. The target's group is initially singleton. Changing the key of $(1, 1)$ to zero creates a tie with one edit, so $\rho_{\text{full}} = 1$. The second ratio is k. □

These elementary families clarify the objective, rather than serve as another priority claim. A whole-table repair optimum cannot simply be reported as this decision radius. They do not exclude reductions between computational problems or the use of general repair/constraint machinery with a different encoding. For $h, k \in \{1, 2, 3\}$, unrestricted raw-destination enumeration checks both costs; together with the coupled example in both edit models, eight checks cover 285,032 destinations. These small illustrative checks are separate from the E008–E011 studies and are not additional source validation.

5 Making a rejected target accept

For row r define the match mask $M(r) = \{j : r_j = 0\}$ and contribution

$$A_{gv}(r) = \mathbf{1}[X_g \subseteq M(r)]\big(\mathbf{1}[r_y = 0] - \mathbf{1}[r_y = v]\big).$$

Since the target contributes one, acceptance means $1 + \sum_{r \neq t} A_{gv}(r) \geq 1$ for all $g, v \neq 0$. Other nonmatching key values can retain their original identities; changing a matching key to nonmatching uses one allowed alternative. A prescribed mask change therefore costs its symmetric-difference size. This predicate quotient is setup rather than a claimed new generic reduction.

Lemma 1 (Canonical accepting actions)**.** *For every original row r and arbitrary destination u, there is a row c with $H(r, c) \leq H(r, u)$ and $A_{gv}(c) \geq A_{gv}(u)$ for every rule and rival. It belongs to the following action set, writing $M_0 = M(r)$:*

1. *if $r_y = 0$: RHS 0 with any supermask of M_0;*
2. *if $r_y \neq 0$: unchanged RHS with any submask of M_0, or RHS 0 with any supermask of M_0.*

Proof. Write $M_* = M(u)$. If $r_y = 0$, choose RHS 0 and mask $M_0 \cup M_*$. Restoring originally matching keys and retaining the original RHS cannot increase cost. Every group containing u also contains c; its contribution is positive instead of zero or negative, and no new negative contribution arises.

If $r_y \neq 0$ and $u_y = r_y$, retain that RHS and choose $M_0 \cap M_*$. Restoring originally nonmatching keys removes edits. Group membership can only shrink relative to u, removing negative contributions and losing no positive one. If instead $u_y \neq r_y$, choose RHS 0 and $M_0 \cup M_*$. Both destinations use one RHS edit. The chosen keys cost no more; expanding positive membership or replacing negative contributions cannot lower a margin. Concrete key values are reconstructed without gratuitous changes between different nonmatching values, so these abstract costs are attainable cell costs. □

Theorem 2 (Exact reduced accepting optimization)**.** *The minimum accepting cost over the canonical actions equals the minimum over all permitted raw cell edits. Each row needs at most* $2^q + 1$ *actions.*

Proof. Apply Lemma 1 to every row of an arbitrary accepting witness. All margins weakly increase and cost weakly decreases, so the resulting canonical table still accepts. Conversely, canonical edits are allowed raw edits. If $p = |M_0|$, an original-zero row has 2^{q-p} actions; an original nonzero row has $2^p + 2^{q-p} \leq 2^q + 1$ actions. □

For action a of row i, use binary z_{ia} with raw cost w_{ia} and contribution $A_{gv}(i, a)$. An exact mixed-integer formulation is

$$\begin{aligned} \min \ & \sum_{i,a} w_{ia} z_{ia}, \\ \text{s.t.} \ & \sum_{a} z_{ia} = 1 \quad \text{for every context row } i, \\ & \sum_{i,a} A_{gv}(i, a) z_{ia} \geq 0 \quad \text{for every } g, v \neq 0. \end{aligned}$$

For fixed keys, restrict each mask to its original value. Rivals absent from a group initially can be omitted in the accepting model: retaining a nonzero RHS only loses matches, while every RHS change goes to zero. Such a rival cannot enter the group, and the protected target ensures a positive target count. This pruning is invalid for rejection, where introducing an absent rival may create a tie.

The reduction is independent of key-domain cardinality, but enumeration remains exponential in q, and the number of rival margins depends on $m(d - 1)$. Neither exactness nor a smaller formulation guarantees a rapid solve.

6 Finite validation and constructed computation

Three locked studies precede the source evaluation. Their protocols were committed before implementations, and implementations before execution. E008 compares canonical-action programs with unrestricted raw-coordinate edits on 5,107 constructed tables and six rule families: all 30,642 radii, 30,618 finite witnesses and 61,854 dominance checks agree. Binary and ternary key/RHS domains include actual changed nonmatching values.

E009 checks Theorem 1 on all 18,779 accepted cases from that finite domain, including 24 impossible one-row cases. All 18,755 finite witnesses verify. These cases overlap E008; they are not independent replications. A separate raw-cell-factorized MILP agrees with raw enumeration in 2,118 cases, including 18 infeasible instances. It uses cell choices and conjunction linearizations, not canonical actions or the formula.

E010 generalizes to 2, 4 and 6 keys. All 162 small raw-oracle fixtures pass before a 72-case constructed comparison with $n \in \{16, 128\}$, key alphabet four, RHS alphabet two/four and two determinant families. Three designed regimes enforce dense acceptance, singleton acceptance or coupled rejection. Each method runs twice with reversed order, rebuilding models each time. SciPy 1.17.1 wraps HiGHS with presolve, zero requested relative gap and a one-second search limit. Total time includes construction and witness decoding; the search limit does not bound complete call time.

Regime	Cases	Both exact twice	Specialized slower
Accepted, dense	24	17	0
Accepted, singleton	24	24	0
Rejected, coupled	24	24	1

Table 2: Complete E010 denominators. Seven dense-accepted generic cases time out in both repetitions; all specialized calls are exact.

All certified optima agree. For rejected cases, the median case-level ratio of generic to canonical median total time is 2.997, conditional on both repetitions being exact. One canonical case is slower: 124.92 versus 81.27 ms. No censored time is substituted into a speedup. The inexpensive accepted formula avoids generic optimization entirely. These are two repetitions on one shared Mac, not hardware-independent or representative timing estimates. The rejected generator contains no target-valued RHS context rows, and the smallest determinant families coincide; these controls limit generalization.

7 Reused-source study

E011 reuses Beers, Flights and Rayyan from the Raha repository [8] at revision `7be1334b`. All source hashes and positional alignment keys are checked. The data and earlier study outcomes had been inspected; this is an exploratory reused-source study, not a blind holdout or Raha-system replication.

Nine fixed tasks use brewery_id to check brewery_name/city/state, flight to check four departure/arrival time columns, and Rayyan's original `jounral_abbreviation` spelling to check journal_issn. Rayyan journal_title jointly uses abbreviation and ISSN determinants. Every target uses its complete dirty table as context. Empty strings are ordinary categories; no unary-format checks apply. Accordingly, these decisions differ from the project's earlier U/Rmode configuration.

There are 18,734 target cells, 13,762 accepted and 4,972 rejected. All accepted full/fixed radii and finite witnesses are computed. The accepted stage is built twice with identical outcome bytes; dirty decisions and witnesses are hash-frozen before reference CSV rows are parsed. Normalization unescapes HTML, collapses runs of ASCII space/tab/newline, and strips those three characters. Reference correctness is normalized dirty/reference equality at the aligned target coordinate, not a semantic adjudication of truth.

Task	Accepted	Fixed ∞ to finite	Finite decrease
Beers brewery_name	2410	127	0
Beers city	2275	127	0
Beers state	2275	127	0
Flights sched_dep_time	1663	0	0
Flights act_dep_time	1133	0	0
Flights sched_arr_time	1411	0	0
Flights act_arr_time	1166	0	0
Rayyan journal_issn	768	384	0
Rayyan journal_title	661	368	1
Total	13762	1133	1

Table 3: Complete accepted-target census. Most key-related reductions change fixed-key impossibility; only one decreases an already finite radius.

No accepted target without singleton groups has a reduction. The sole finite decrease is Rayyan row 644, journal_title, with groups of sizes four and one. Its fixed radius is two. Changing journal_issn at context row one to the target's value creates a singleton-group rival, giving full radius one. The target is reference-correct. This is a post-run illustrative selection from the complete census, not a separate confirmatory example.

For rejected targets, a dirty-only hash orders row coordinates within each task; the first four are selected, or all if fewer. This yields 32 targets because Beers brewery_name has no rejected cell. Each full/fixed canonical MILP runs once with a five-second search limit, without warm starts or retries. All 32 fixed solves are optimal; 30 full solves are optimal, and every exact pair has equal cost. Thus the declared prediction of at least one strict gain in the selected joint Rayyan task is unsupported. Its exact pairs, rows 271 and 138, both have cost 104. Full models for rows 509 and 146 time out without a primal or bound; their fixed optima are 104. Each has 4,774 variables, 1,396 constraints and 397 active rival margins. Their full optima remain unknown.

Among accepted targets, 1,816 differ from their references; 1,493 of these have full radius at least two. Hence stable acceptance can disagree with a reference. These finite counts are not independent trials, calibrated error probabilities or an accuracy improvement. The rejected subset is not a representative census.

8 Related work and scope

Cell-level repair and intervention objectives substantially predate this work. Debosschere and Geerts [1], Section 2, define cell modifications, their cardinality distance and repairs satisfying the FD set. Their Propositions 1–3 concern responsibility complexity and repair/cause correspondence. Our edited table may violate FDs: the goal is a protected target's plurality-decision flip, not consistency restoration. Proposition 1 makes the numerical distinction explicit. Bertossi [2], Sections 3.1 and 4, gives abstract repair semantics and an attribute-cause instantiation using NULL replacements. We instead allow arbitrary domain values, do not assign causal responsibility to a particular cell, and need not preserve other outputs.

Meliou et al. [3], Sections 4.1–4.4 and Theorems 14, 15 and 17, give predicate partitions, cause/responsibility equivalence and SAT-based reasoning about changed inputs. General vote operations already optimize constructive/destructive goals with score constraints [4], Definitions 2–3 and Section 4. Those generic frameworks motivate an encoding; they are not claimed as new here. Our accepting proof additionally constructs a no-more-expensive row destination dominating

every relevant margin, and our rejection proof constructs a witness attaining the margin/import lower bound. These are the specialized statements being presented; their absence from the inspected theorem statements is not a systematic originality certificate.

FD repairs with immutable trusted cells are established [5]. Certified k-NN over FD repairs [6] uses different possible worlds and edit semantics. Recent work distinguishes AFD strength from detection utility [7]; our radius likewise must not be interpreted as accuracy. The practical evidence here concerns sensitivity and bounded computation, not a demonstrated improvement to a deployed auditing workflow.

The same-RHS restriction permits a single change to zero to improve all relevant margins. Different dependents, weighted costs, additional constraints or mutable targets can invalidate the proofs. Empty-as-category source semantics are deliberate, and reference labels may themselves be imperfect. Source reuse, small synthetic sizes, sparse rejected sampling, two solver timeouts and the lack of independent scientific assessment limit the empirical conclusions.

9 Reproducibility and provenance

The frozen M31 companion preserves the original E008–E011 protocols, code, source manifests, all outcomes, witnesses and separate delivery checks. Its manuscript snapshot predates this revision. A separate R2 supplement provides the raw enumerator and all records for the new analytical examples, the comparison notes and revision verification. The original source CSV cache was unavailable at this revision; no new raw-source reconstruction or solver/timing run is claimed. No public code URL or arXiv submission is claimed in this working draft. E008 protocol/implementation commits are `efc1365/14fe214`; E009 uses `96962be/05a2b07`; E010 uses `fbd04cb/2212b58`; E011 uses `86dad7d/fd341d4`. The exact environment for E009–E011 is Python 3.13.5, NumPy 2.1.3 and SciPy 1.17.1 on an arm64 Mac. Solver-optimal results require a checked integer-feasible witness, objective agreement and bound/gap checks within 10^{-7}. Time-limit outcomes stay censored.

E011's separate checker uses original categorical codes, direct formulas and incremental group-count updates, rechecking 26,453 finite source witnesses, selection hashes, reference labels and stage hashes. Such separate code is self-review in one research environment, not independent human verification. Source cell strings remain outside the outputs; witnesses contain original row/column coordinates and target-relative destination codes. Exact radius histograms and reference buckets are retained, including all negative outcomes. Generative AI was used substantially for ideation, proofs, code and writing. The named human author's substantive review is pending; this draft is not authorization to submit or publish it.

10 Conclusion

Editable keys have a precisely characterized effect on accepted-target stability: only singleton groups can improve on the fixed-key rejection cost. For accepting counterfactuals, canonical row dominance preserves exact optima while reducing raw categorical choices. The implementations have finite validation and bounded computational value, but reused-source evidence is concentrated in fixed-infinite transitions and does not show a rejected-subset cost gain. Reporting these limits keeps decision stability distinct from correctness and the specialized result distinct from generic intervention ideas.

References

[1] M. Debosschere and F. Geerts. Cell-Based Causality for Data Repairs. TaPP, 2015. USENIX proceedings.

[2] L. Bertossi. Specifying and Computing Causes for Query Answers in Databases via Database Repairs and Repair Programs. arXiv:1712.01001v7, 2020.

[3] A. Meliou, W. Gatterbauer, S. Nath and D. Suciu. Tracing Data Errors with View-Conditioned Causality. SIGMOD, 2011. doi:10.1145/1989323.1989376.

[4] L. Xia. How Many Vote Operations Are Needed to Manipulate a Voting System? 2012. arXiv:1204.1231v3.

[5] G. Beskales, I. F. Ilyas and L. Golab. Sampling the Repairs of Functional Dependency Violations under Hard Constraints. PVLDB 3(1), 197–207, 2010. doi:10.14778/1920841.1920870.

[6] A. Z. Fan and P. Koutris. Certifiable Robustness for Nearest Neighbor Classifiers. ICDT, 2022. doi:10.4230/LIPIcs.ICDT.2022.6.

[7] X. Wan and X. Han. Strength Is Not Value: Rethinking Approximate Functional Dependency Measures for Error Detection. 2026. arXiv:2610.08189v1.

[8] M. Mahdavi, Z. Abedjan, R. Castro Fernandez, S. Madden, M. Ouzzani, M. Stonebraker and N. Tang. Raha: A Configuration-Free Error Detection System. SIGMOD, 2019. doi:10.1145/3299869.3324956.